

CURRICULUM VITAE: ALESSANDRO LICCIARDI



Contact Information

Address: Department of Mathematical Sciences (DISMA)
Politecnico di Torino
Corso Duca degli Abruzzi 24, 10129, Torino, Italy
Email: alessandro.licciardi@polito.it
Website: <https://alelicciardi99.github.io/alicciardi>
GitHub: <https://github.com/alelicciardi99>
LinkedIn: <https://linkedin.com/in/alessandro-licciardi>
ORCID: 0009-0006-8428-3408

Personal Information

Nationality: Italian
Languages: Italian: Native
English: Native

Current Position

PhD Student in Mathematical Sciences, DISMA, Politecnico di Torino (since November 2023)
Principal Supervisor: Prof. Lamberto Rondoni

Research Visits

Visiting Student Researcher, Rotskoff Lab, Department of Chemistry, Stanford University, Stanford, CA. 2025 – 2026
Host: Prof. Grant M. Rotskoff

Research Interests

Machine learning theory, preprocessing operators, federated learning, and generative models, with cross-disciplinary applications in healthcare, physics, and natural sciences.

Academic Formation

2023 - ongoing	PhD in Mathematical Sciences Department of Mathematical Sciences (DISMA), Politecnico di Torino, Torino (Italy) Supervisor: Prof. Lamberto Rondoni
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2021 - 2023	Master of Science in Mathematical Engineering , 110/110 <i>cum Laude</i> DISMA, Politecnico di Torino, Torino (Italy) Thesis: <i>Wavelet Scattering Transform. Mathematical Analysis and Applications to VIRGO Gravitational Waves Data</i> Supervisor: Prof. Lamberto Rondoni
2018 - 2021	Bachelor of Science in Mathematics for Engineering , 110/110 <i>cum Laude</i> DISMA, Politecnico di Torino, Torino (Italy)

Scholarships and Awards

2023 - 2026	Industrial PhD Scholarship (MUR DM 117), co-funded by the Italian Ministry of University and Research and Eltek S.p.A. Politecnico di Torino, Torino (Italy)
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List of Publications

Journal Articles

1. A. Licciardi, D. Leo, and D. Carbone. *Wavelet Scattering Transform and Fourier Representation for Offline Detection of Malicious Clients in Federated Learning*. Accepted in: IEEE Internet of Things Journal, 2026. <https://doi.org/10.1109/JIOT.2026.3671698>
2. A. Licciardi, S. Bernardi, M. Pizzi, P. Begnamino, and L. Rondoni. *A Machine Learning Model for Humidity Estimation Based on Physics-Informed Dimensionality Reduction*. In: Nonlinear Dynamics, 113(23), 31941–31963, 2025.
3. A. Licciardi, D. Carbone, L. Rondoni, and A. Nagar. *Wavelet Scattering Transform for Gravitational Waves Analysis: An Application to Glitch Characterization*. In: Physical Review D, 111(8), 084044, 2025. <https://doi.org/10.1103/PhysRevD.111.084044>
4. A. Licciardi and D. Carbone. *WhaleNet: A Novel Deep Learning Architecture for Marine Mammals Vocalizations on the Watkins Marine Mammal Sound Database*. In: IEEE Access, 12, 154182–154194, 2024. <https://doi.org/10.1109/ACCESS.2024.3482117>

Conference Proceedings

1. A. Licciardi, D. Leo, E. Fani, B. Caputo, and M. Ciccone. *Interaction-Aware Gaussian Weighting for Clustered Federated Learning*. In: Proceedings of the 42nd International Conference on Machine Learning (ICML), PMLR 267:37642–37666, 2025.
2. A. Licciardi. *WASA: Wavelet Scattering Autoencoders for Unsupervised Offline Detection of Malicious Clients in Federated Learning*. to appear in IEEE International Conference on Omni-layer Intelligent Systems (COINS), 2026.
3. A. Licciardi, D. Carbone, and L. Rondoni. *Wavelet Scattering Operators for Multiscale Processes: The Case Study of Marine Mammal Vocalizations*. In: Proceedings of the 2nd International Conference on Nonlinear Dynamics and Applications (ICNDA), pp. 173–191. Springer Nature Switzerland, 2024. https://doi.org/10.1007/978-3-031-69146-1_15

Under Review

1. A. Licciardi, R. Raineri, A. Proskurnikov, L. Rondoni, and L. Zino. *Tackling Heterogeneity in Federated Learning via Variance-Reduced Boltzmann Sampling within Homogeneous Social Coalitions*. Submitted to: IEEE Internet of Things Journal, 2026.
2. L. Deriu and A. Licciardi. *Malicious Client Detection via Wavelet Scattering Autoencoders in Federated Learning for Coffee Leaf Disease Classification*. Submitted to: Elsevier Ecological Informatics, 2026.
3. B. Düring, M. Fraia, and A. Licciardi. *From Graphons to Real-World Networks: Kinetic Opinion Dynamics under Selective Media Influence*. Submitted to: Mathematical Models and Methods in Applied Sciences (M3AS), 2026. <https://arxiv.org/abs/2607.02821>

Peer-Review Activities

- Invited Reviewer for **Conference on Neural Information Processing Systems (NeurIPS)**, 2026.
- Reviewer for **International Conference on Learning Representations (ICLR)**, 2026.
- Referee for **IEEE Journal of Oceanic Engineering**, 2025–present. (2 manuscript reviewed)
- Referee for **Ecological Informatics** (Elsevier), 2025–present. (6 manuscripts reviewed)

Teaching

- 2025–present: **Teaching Assistant**, Politecnico di Torino, Torino, Italy. Lectures in *Mathematical Analysis II* (English and Italian courses).
- 2024–present: **Teaching Assistant**, Politecnico di Torino, Torino, Italy. Lectures in *Mathematical Analysis I* (English course).

Students

- 2025, **Simone Parisi**, master's student in Physics of Complex Systems at Università degli Studi di Torino.
Title of the thesis: *Inverse Problems with Diffusion Models and Physics-Informed Machine Learning*. *In progress*.
- 2026, **Andrea Rinaudo**, master's student in Mathematical Engineering at Politecnico di Torino.
Title of the thesis: *Security Analysis of Federated Split Learning Protocols*. *In progress*.

Participation in Research Groups

Since 2024: Associated Member of **Istituto Nazionale di Alta Matematica (INdAM)**, Gruppo Nazionale di Fisica Matematica (GNFM).

Since 2023: Member of **Models and Methods of Mathematical Physics** research group, DISMA, Politecnico di Torino. PI: Prof. Lamberto Rondoni.

Since 2023: Associated Member of **Istituto Nazionale di Fisica Nucleare (INFN)**, Sezione di Torino.

Conferences, Workshops and Talks

Seminars of the Department of Mathematics, 2 July 2026, Università di Bologna, Bologna (Italy).

Invited talk: *Mathematical Advances in Federated Learning: Interaction Modeling and Clustering, Spectral Representations and Anomaly Detection*

Cambridge Machine Learning Systems Seminar (CaMLSys), 15 June 2026, Computer Laboratory, University of Cambridge (UK).

Invited talk: *Principled, Communication-Light Federated Learning: Interaction Modeling and Offline Spectral Representations for Clustering and Robustness*

Italian Conference on Computational Social Science (CS2Italy), 19–21 May 2026, Torino (Italy).

Poster: *Tackling Heterogeneity in Federated Learning via Variance-Reduced Boltzmann Sampling within Homogeneous Social Coalitions*

ScatterBrains, 16 April 2025, Politecnico di Torino, Torino (Italy).

Contributed talk: *Applications of the Wavelet Scattering Transform in Machine Learning: from Gravitational Waves and Whale Vocalizations to Communication-Efficient Anomaly Detection in Federated Learning*

42nd International Conference on Machine Learning (ICML), 13–19 July 2025, Vancouver (Canada).

Poster and accepted paper: *Interaction-Aware Gaussian Weighting for Clustered Federated Learning*

GNFM Summer School in Mathematical Physics, September 2024.

Contributed talk: *Wavelet Scattering Operators for Machine Learning in Bioacoustics and Astrophysics*

GIMC-SIMAI Young, July 2024, Università degli Studi Federico II, Napoli (Italy).

Invited speaker. Contributed talk: *Non-Linear Operators in AI: from Time-Series Analysis to Texture Discrimination*

Generative Modeling Summer School (GeMSS), June 2024, Eindhoven University of Technology, Eindhoven (Netherlands).

Poster: *Non-Linear Pre-Processing Operators for Signal Classification and Time Series Analysis*

2nd International Conference on Nonlinear Dynamics and Applications (ICNDA), February 2024, Sikkim Manipal University (India).

Contributed paper and talk: *Wavelet Scattering Operators for Multiscale Processes: The Case Study of Marine Mammal Vocalizations*

Skills

Programming Languages: Python, Matlab, R, C/C++, SQL, L^AT_EX

Machine Learning and Statistics: Federated Learning, Wavelet Scattering Transform, Generative Models, Deep Neural Networks, Preprocessing Operators, Time Series Analysis, Dimensionality Reduction, Physics-Informed Machine Learning

Libraries and Software: PyTorch, Scikit-learn, SciPy, Kymatio, Pandas, NumPy, Statsmod-

els, JAX, GwPy, JAGS

Platforms: Windows, Linux

Other Interests and Hobbies

Scientific dissemination, music (electric and acoustic guitar: psychedelic rock, blues, jazz), writing and poetry